Handout: Feasibility of the learning problem

Oriol Pujol

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1 Previous Definitions

- Domain set: An arbitrary set $\mathcal{X}$ where data belongs.
- Label set: The set $\mathcal{Y}$ that denotes the output set for a certain task. Same data may lead to different tasks by just changing this label set. In the most simple case, this is a binary set $\mathcal{Y} = \{0, 1\}$.
- Training data / Observed samples: We consider a set $\mathcal{S} = \{(x_1, y_1) \dots (x_m, y_m)\}$ a finite set of m elements in $\mathcal{X} \times \mathcal{Y}$.
- Data generation process: Over the domain set $\mathcal{S}$ we may assume that there is a distribution $\mathcal{D}$ that denotes the probability of each element in the set being generated. As such, any observed data is $x \in S|_x$ is drawn from that distribution as $x \sim D$, where $S|_x$ corresponds to the x part of the elements in S. As a first approximation during the begining of this text we assume that there exists a deterministic function f that takes care of the y part of $\mathcal{S}$, defined as $f : \mathcal{X} \to \mathcal{Y}$. With this, every $y_i = f(x_i)$. Later we will consider the case in which the distribution D is defined over $\mathcal{X} \times \mathcal{Y}$.
- We call hypothesis, h, to an element of a set in the hypotheses space or model class $\mathcal{H}$ that defines the family of functions or, in other words, the functional form of the prediction. Thus, given a specific $\mathcal{H} \subset Hom(\mathcal{X}, \mathcal{Y})$ (homeomorphism from $\mathcal{X}$ to $\mathcal{Y}$), $h \in \mathcal{H}$ our prediction is $h(x_i)$.
- Measure of success: The first measure of success we can think of is the probability of error. This is the probability of not correctly classify. We can generalize this concept by considering the notion of error in a more general setting, like for example, multi-class prediction, or real valued prediction (regression). We encode this notion of mistake or more precisely mismatch in the concept of **loss**, this is what is the effect of the misprediction.

  Consider a domain set Z (for example $\mathcal{X} \times \mathcal{Y}$), and a set of models of the universe M (where $\mathcal{H} \subseteq M$), and a notion of a loss of some model on a given instance, $\ell(h, z)$, $h \in M$. The data is generated by some unknown

distribution D over Z, we aim to find the best model for that distribution. So we define the **risk** or **generalization loss** as,

$$L_D(h) = \mathbb{E}_{z\sim D}\ell(h,z).$$

Observe that, if $Z = X \times \{0,1\}$ and M is a set of functions from $f : X \to \{0,1\}$, and the loss we used was

$$\ell(h,(x,y)) = \mathbb{1}[h(x) \neq y] = \begin{cases} 1 & \text{if } h(x) \neq y, \\ 0 & \text{if } h(x) = y. \end{cases}$$

then,

$$L_D(h) = \mathbb{E}_{(x,y)\sim D}\mathbb{1}[h(x) \neq y] = P_{(x,y)\sim D}[h(x) \neq y],$$

which is the probability of error. And we recover our intuition about the measurement.

In the regression case, $Z = X \times \mathbb{R}$ and M is a set of functions from $f \to \mathbb{R}$, the loss would be encoding the notion of big/small mistake, as for example, the square loss,

$$\ell(h,(x,y)) = (h(x) - y)^2.$$

- Empirical risk (or training error): We can use the same notion applied to the data set S. In this case we define,

$$L_S(h) = \frac{1}{|S|}\sum_{z\in S}\ell(h,z).$$

Observe that in the case that the loss function is defined as $\mathbb{1}[h(x) \neq y]$ this is

$$L_S(h) = \frac{1}{m}\sum_{i=1}^{m}\mathbb{1}[h(x_i) \neq y_i] = \frac{|\{i \in [m] : h(x_i) \neq f(x_i)\}|}{|S|},$$

where $[m] = \{1 \dots m\}$ and $|\cdot|$ is the cardinality of the set. Note that $|S| = m$.

2 Algorithms and Empirical Risk Minimization

Definition 2.1. *(Algorithm) We define a learner or algorithm as*

$$A : \bigcup_{m=1}^{\infty}(X \times \{0,1\})^m \to \{f | f : X \to \{0,1\}\}.$$

It takes as input any set of samples of any size m and outputs a set of functions f that predict that set of samples.

Definition 2.2. *(Empirical Risk Minimization / ERM). Given a model class $\mathcal{H}$ and a sample $\mathcal{S}$, the empirical risk minimization is an algorithm described as*

$$h_S \in \arg\min_{h \in \mathcal{H}} L_S(h). \tag{1}$$

3 Learning under the realizability assumption

We start assuming that there exists a deterministic function f that takes care of the y part of $\mathcal{S}$, defined as $f : \mathcal{X} \to \mathcal{Y}$. With this, every $y_i = f(x_i)$. We, additionally, consider the following (hard) assumption,

Assumption 1. *(Realizability Assumption) There exists $h^* \in \mathcal{H}$ s.t.*

$$L_{D,f}(h^*) = 0.$$

Because all the elements in S are drawn from D, this implies that $L_S(h^*) = 0$. And because of Eq. 1, then there exists sample sets S such that $L_S(h_S) = 0$.

Formally, our goal is to upper bound ($< \delta$) the probability of $L_{D,f}(h_S) > \epsilon$, this is the probability of having a sample set that leads to an h_S such that it displays bad generalization error ($> \epsilon$) but looks good $L_S(h_S) = 0$,

$$\mathcal{D}^m(\{S|_x : L_{D,f}(h_S) > \epsilon\}) < \delta,$$

where $S|_x$ correspond to the x component of S. I am using the following notation here, $\mathcal{D}^m(\cdot) = P_{S \sim \mathcal{D}^m}(\cdot)$, with the exponent m denoting the cardinality of the sample set S.

Let us consider the set of "bad" hypotheses, i.e.

$$\mathcal{H}_B = \{h \in \mathcal{H} : L_{D,f}(h) > \epsilon\},$$

and the samples that mislead the predictor

$$M = \{S|_x : \exists\, h \in \mathcal{H}_B, L_S(h) = 0\}.$$

Observe that this set is the set of all sample sets that induce a bad hypothesis but they looks good from the point of view of the empirical error. Observe that

$$\{S|_x : L_{D,f}(h_S) > \epsilon\} \subseteq M.$$

M can be rewritten as

$$M = \bigcup_{h \in \mathcal{H}_B} \{S|_x : L_S(h) = 0\}.$$

Thus,

$$\mathcal{D}^m(\{S|_x : L_{D,f}(h_S) > \epsilon\}) \leq \mathcal{D}^m(M) = \mathcal{D}^m(\bigcup_{h \in \mathcal{H}_B} \{S|_x : L_S(h) = 0\}).$$

(Recall that $h_S \in \mathcal{H}_B$.)
We will use the union bound ($P(A \bigcup B) \leq P(A) + P(B)$),

$$\mathcal{D}^m(\{S|_x : L_{D,f}(h_S) > \epsilon\}) \leq \sum_{h \in \mathcal{H}_B} \mathcal{D}^m(\{S|_x : L_S(h) = 0\}). \tag{2}$$

Now let us bound $\mathcal{D}^m(\{S|_x : L_S(h) = 0\})$.
Using the i.i.d. assumption

$$\mathcal{D}^m(\{S|_x : L_S(h) = 0\}) = \prod_{i=1}^{m} \mathcal{D}(\{x_i : h(x_i) = f(x_i)\}). \tag{3}$$

Observe that $L_S(h) = 0$ implies that $h(x_i) = f(x_i)$ for all m samples.

$$\mathcal{D}(\{x_i : h(x_i) = f(x_i)\}) = 1 - L_{D,f}(h) \leq 1 - \epsilon,$$

where the probability of having the hypothesis right is $1 - L_{D,f}(h)$. Remember that $L_{D,f}(h)$ is the probability of error. Now, we consider the condition that $L_{D,f}(h) > \epsilon$ and we get the last bit of the former inequality. Combining this with Eq.3 we obtain,

$$\mathcal{D}^m(\{S|_x : L_S(h) = 0\}) \leq (1 - \epsilon)^m.$$

And, using the result of Eq.2 we have,

$$\mathcal{D}^m(\{S|_x : L_{D,f}(h_S) > \epsilon\}) \leq \sum_{h \in \mathcal{H}_B} \mathcal{D}^m(\{S|_x : L_S(h) = 0\}) \leq |\mathcal{H}|(1 - \epsilon)^m. \tag{4}$$

We can express it in nicer way by considering the definition of the exponential function $e^x = \sum_i x^i/i!$. Thus, $e^{-\epsilon} = 1 - \epsilon + \epsilon^2/2 - ....$ And we can bound $1 - \epsilon \leq e^{-\epsilon}$. Combining this with Eq.4 we obtain,

$$\mathcal{D}^m(\{S|_x : L_{D,f}(h_S) > \epsilon\}) \leq \sum_{h \in \mathcal{H}_B} \mathcal{D}^m(\{S|_x : L_S(h) = 0\}) \leq |\mathcal{H}|e^{-\epsilon m}. \tag{5}$$

Thus, observe that if $\delta = |\mathcal{H}|e^{-\epsilon m}$, then for

$$m \geq \frac{\log(|\mathcal{H}|/\delta)}{\epsilon},$$

and if the realizability assumption holds, then with probability $1-\delta$ over the choice of i.i.d. for any f and any distribution $\mathcal{D}$ we have that for every $ERM_{\mathcal{H}}$ hypothesis h_S, it holds that

$$L_{D,f}(h_S) \leq \epsilon.$$

This is, for sufficiently large m, the solution of $ERM_{\mathcal{H}}$ will be *probably* (with confidence $1-\delta$) *approximately* (up to a confidence ϵ) *correct*. This is PAC learning.

4 PAC-learning theorems

Definition 4.1. *We say that the class of predictors H is PAC (Probably Approximately Correct) learnable if there exists a function*

$$m_H : \underbrace{(0,1)}_{accuracy/precission(\epsilon)} \times \underbrace{(0,1)}_{confidence(\delta)} \to \mathbf{N}$$

such that there exists a learner (algorithm)

$$A : \bigcup_{m=1}^{\infty} (X \times \{0,1\})^m \to \{f | f : X \to \{0,1\}\}$$

that takes as input any set of samples of any size m and outputs a set of functions f that predict that set of samples. Thus, given $S \sim D^m$ then $A(S)$ is an algorithm that outputs a function h, that can be or not be from H. Then, there exists that learner, A that for every D a probability distribution over X, every $f \in H$ (labelling function), and $\epsilon, \delta > 0$,

$$P_{S \sim D^m, f}[L_{D,f}(A(S)) > \epsilon] < \delta, \quad \forall m \geq m_H(\epsilon, \delta).$$

Observe that m_H is the smallest number of points for successful learning.

Theorem 4.2. *Any finite H is PAC learnable with*

$$m_H(\epsilon, \delta) \leq \frac{\ln |H| + \ln (1/\delta)}{\epsilon}.$$

Proof. As per previous Section. □

Lemma 4.3. *Under the conditions of Theorem 4.2, any ERM_H learning will be successful.*

5 Generalization to Distributions

The generalization we are going to use is that data is generated by some probability distribution D over $X \times Y$. Thus labels are random.

The best rule we can produce is **Bayes rule**:

$$h^*(x) = \begin{cases} 1 & \text{if } D((x,1)|x) \geq 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

This is probability of observing label 1 observing x. But we do not know D we only see a sample generated by D. The whole problem of learning is that there is a distribution and we only observe a sample S from that distribution.

6 Agnostic PAC-learning

Now we redefine successful learning to have only a *relative* error guarantee. It is relative compared to the best possible solution belonging to the hypothesis class. Given a hypothesis class H, the best solution is given by $\min_{h' \in H} L_D(h')$. Our notion of success becomes ϵ-success, i.e

$$L_D(h) \leq \min_{h \in H}(L_D(h')) + \epsilon.$$

This is, we consider our hypothesis h successful as long as its error deviates from the best error achievable using that model family by a quantity, at most, ϵ. The notion of learnability, will proceed similar to the former case, we will be looking for $m_H(\epsilon, \delta)$ that will tell us how many samples we do need for ϵ-success.

Definition 6.1. *Definition of agnostic PAC-learnability. We say that the class of predictors H is agnostic PAC learnable if there exists some function*

$$m_H(\epsilon, \delta) : (0,1) \times (0,1) \to \mathbb{N}$$

such that there exists a learner (algorithm)

$$A : \bigcup_{m=1}^{\infty} (X \times \{0,1\})^m \to \{f | f : X \to \{0,1\}\}$$

such that for every D a probability distribution over $X \times Y$, and $\epsilon, \delta > 0$,

$$P_{S \sim D^m}[L_D(A(S)) > \min_{h \in H}(L_D(h)) + \epsilon] < \delta, \quad \forall m \geq m_H(\epsilon, \delta).$$

6.1 Uniform Convergence

If we look at the definition of agnostic PAC-learning, we want to bound the generalization error relative to the best generalization error that we can achieve in the model class. However, we need to relate this to the empirical error. Uniform convergence is a strategy to ensure that empirical risk (remember that given a sample S, ERM outputs the hypothesis $h \in H$ that achieves the minimum risk on the sample) is related to the generalization risk.

We stat with how well the samples represent the distribution.

Definition 6.2. *A sample $S = (z_1, \ldots, z_m))$ (or, alternatively $S = \{(x_1, y_1) \ldots (x_m, y_m)\}$ is ϵ-representative of a class H w.r.t the distribution D if*

$$\forall h \in H \quad |L_S(h) - L_D(h)| \leq \epsilon,$$

We are relating the empirical loss, $L_S(h)$, to the generalization loss, $L_D(h)$.

Observation 1. *Note that if S is indeed representative of H w.r.t. D then ERM_H is a good learning strategy. Remember that ERM_H is picking a member of H that minimizes the empirical risk.*

Lemma 6.3. *If S is ϵ-representative of H w.r.t. D then for any ERM_H function h_S*

$$L_D(h_S) \leq \min_{h \in H} L_D(h) + 2\epsilon$$

Proof.

$$L_D(h_S) \leq L_S(h_S) + \epsilon \leq \min_{h \in H} L_S(h) + \epsilon$$

$$\min_{h \in H} L_S(h) + \epsilon \leq \min_{h \in H} (L_D(h) + \epsilon) + \epsilon$$

□

Definition 6.4. *Let us define the sample complexity of uniform convergence w.r.t. H, $m_H^{UC}(\epsilon, \delta)$, as the minimum number m such that for every distribution D if we pick a sample of size m, $S \sim D^m$, then with probability of at least $1 - \delta$, then S is ϵ-representative.*

If we have uniform convergence (enough samples $m_H^{UC}(\epsilon, \delta)$) then with high probability our sample is ϵ-representative, with that and Lemma 6.3 then ERM will work. Then, it will be agnostic PAC learnable.

Corollary 6.4.1. *The sample complexity of agnostic PAC learnable relates to the sample complexity of uniform convergence as follows,*

$$m_H(\epsilon, \delta) \leq m_H^{UC}(\frac{\epsilon}{2}, \delta).$$

6.2 Learning under uniform convergence

We start recalling this classical result from concentration of measure.

Theorem 6.5. *(***Hoeffding's inequality***) Assume $\theta_1, \theta_2, , \theta_m$ are i.i.d. random variables, with mean μ that take values in $[a, b]$, then*

$$P(|\mu - \frac{1}{m} \sum_i \theta_i| > \epsilon) \leq 2e^{\frac{-2\epsilon^2 m}{(b-a)^2}}.$$

Now we proceed to bound the probability of a sample not being ϵ-representative **for a single given** $h \in H$,

$$P(|L_D(h) - L_S(h)| > \epsilon) = P(|\mathbb{E}_{z\sim D}[l(h,z)] - \frac{1}{m}\sum_{z\in S} l(h,z_i)| > \epsilon).$$

Using Hoeffding's for the 0-1 loss $[a,b] = [0,1]$, we have,

$$P(|\mathbb{E}_{z\sim D}[l(h,z)] - \frac{1}{m}\sum_{z\in S} l(h,z_i)| > \epsilon) \leq 2e^{-2\epsilon^2 m}.$$

This equation stands for a single h. However, Uniform Convergence considers ϵ-representative samples. And by Def. 6.2 an ϵ-representative sample applies $\forall h \in H$. So our goal is to bound the probability of a sample S not being ϵ-representative. This is, there exist some $h \in H$ for which ϵ-representativeness does not hold. This is

$$P(\text{S is not eps rep w.r.t some } h \in H) = P[\ \exists h \in H., s.t.\ |L_D(h) - L_S(h)| > \epsilon]).$$

The probability that there is at least one h that fails is upper-bounded by the probability that all of them fail (this includes the culprits that fail).

$$P(\ \exists h \in H., s.t.\ |L_D(h) - L_S(h)| > \epsilon]) \leq \sum_{h\in H} P[|L_D(h) - L_S(h)| > \epsilon]) \leq$$

$$\leq \sum_{h\in H} 2e^{-2\epsilon^2 m} =$$

$$= 2|H|e^{-2\epsilon^2 m}$$

Thus,

$$m_H^{UC}(\epsilon,\delta) \leq \frac{\log(\frac{2|H|}{\delta})}{2\epsilon^2}.$$

Corollary 6.5.1. *The sample complexity of agnostic PAC learning is*

$$m_H(\epsilon,\delta) \leq \frac{2\log(\frac{2|H|}{\delta})}{\epsilon^2}.$$

We observe that this holds for any finite hypothesis class. What about infinite classes? If we are pragmatic enough we will acknowledge that those models are realized in a computer. In that case, the values of the parameters are forceably discrete and bounded to the computational representation of the real values. In summary, any hypothesis class in a computer is pragmatically finite, and as such it is agnostic PAC-learnable.

Corollary 6.5.2. *(The discretization trick) Consider that the hypothesis class has d parameters and they are represented by z bits, let's say $z = 64$. then there are at most $|H| \leq 2^{d \cdot z}$ different hypothesis in H. From corollary 6.5.1 we can say that the sample complexity is*

$$m_H(\epsilon, \delta) \leq \frac{2\log(\frac{2 \cdot 2^{zd}}{\delta})}{\epsilon^2} \leq \frac{2zd + 2\log(\frac{2}{\delta})}{\epsilon^2} \leq \frac{128d + 2\log(\frac{2}{\delta})}{\epsilon^2}.$$

6.3 Formal definition of overfitting

Definition 6.6. *Given an algorithm A and a sample set $\mathcal{S}$, we say that A* **overfits** *(at sample size m) if there exist $\varepsilon, \delta > 0$ such that*

$$P_{S \sim \mathcal{D}^m}(|L_D(A(S)) - L_S(A(S))| > \varepsilon]) \geq \delta$$

This is, with non-negligible probability over the draw of the sample, the hypothesis returned by A has much lower training loss than true loss (a large positive generalization gap).

7 Infinite classes and The No-Free Lunch Theorem

Theorem 7.1. *Let χ be a domain of size n, ($|\chi| = n$). Let H_n^{all} be the set of all possible labellings in that domain, i.e.*

$$H_n^{all} = \left\{h : \chi \to \{0, 1\}\right\}$$

then, the sample complexity of the realizable case is,

$$m_{H_n^{all}}(\frac{1}{8}, \frac{1}{7}) \geq \frac{n}{2}.$$

This means if we see half of the domain (samples) with probability 1/7th we will get error 1/8th.

Remark: $|H_n^{all}| = 2^n$ (because it is finite, this is PAC-learnable under the conditions stated in the former theorem).

Theorem 7.2. *(No-free lunch theorem) Let A be any learning algorithm for a binary classification with 0-1 loss over a domain χ. Let m be any number smaller than $|\chi|/2$ representing the training set size. Then, there exists a distribution over $\chi \times \{0, 1\}$ such that:*

1. *There exists a function $f : \chi \to \{0, 1\}$ with $L_D(f) = 0$ (The function is H_n^{all}),*
2. *With probability of at least $1/7$ over the choice of $S \sim D^m$ we have that $L_D(A(S)) \geq 1/8$ (to be learnable, we need $m > |\chi|/2$).*

Corollary 7.2.1. *The class of all labelling functions over an infinite domain is not PAC-learnable.*

$$m_{H_\infty^{all}}(\frac{1}{8}, \frac{1}{7}) \geq \infty.$$

Observation 2. *We need to constraint the hypothesis class so not all possible labellings are possible. This is introducing an* **inductive bias**. *This is done by selecting a specific hypothesis class.*

Observation 3. *The no free-lunch theorem states that if no inductive bias is given over the hypothesis class (it may display all possible solutions), any algorithm that operates over the hypothesis class of all possible labelings will fail on some learning task.*

7.1 Bias-variance trade-off

Recalling the agnostic PAC-learning setting, we are bounding the term

$$P\big(L_D(h) - \min_{h' \in \mathcal{H}} L_D(h') > \epsilon\big).$$

In this, we can identify the contribution of two elements in

$$L_D(h) = \varepsilon_{app} + \varepsilon_{est},$$

corresponding to

$$\varepsilon_{app} = \min_{h' \in \mathcal{H}} L_D(h') \quad \text{and} \quad \varepsilon_{est} = L_D(h) - \varepsilon_{app}.$$

These two terms correspond to:

- **Approximation error (ε_{app}):** The minimum value achievable in the hypothesis class. It measures the risk we have because we selected that hypothesis class. This can be considered as a measure of **inductive bias** as it does not depend on the sample size. A larger hypothesis class may reduce this value.
- **Estimation error (ε_{est}):** This term identifies how far we are from the minimum achievable. This is, because we are in front of an estimate of the true risk. The quality of this estimate depends on the number of samples (decreases with m) and the size of the hypothesis class (increases logarithmically with $|H|$).

This defines the bias-complexity trade-off.

7.2 VC-dimension

We need a tool to know if a a problem is learnable by a model in the PAC sense even in front of infinite classes. This is the Vapnik-Chervonenskis's dimension (VC dimension). In order to define it we need to first define the concept of *shattering*.

Definition 7.3. *Let H be a class of $\{0,1\}$ functions over some domain X, and let $A \subseteq X$, we say that H shatters A if*

$$\forall g : A \to \{0,1\},\ \exists h \in H \mid \forall x \in A, h(x) = g(x)$$

This means that given any particular set of data points (A in general position) labelled (by g), there exists a function in our hypothesis class H that correctly classifies all the points in A.

Definition 7.4. *The VC-dimension*

$$VCdim(H) \stackrel{def}{=} \begin{cases} \max\{|A| : A \text{ is shattered by } H\}, \\ \infty \text{ if arbitrarily large } A \text{ is shattered.} \end{cases}$$

Observation 4.

$$VC_{dim}(H) \leq \log_2(|H|)$$

which means that $|H| \geq 2^{VC_{dim}(H)}$, because it takes $2^{|A|}$ hypotheses to shatter a set A since I have to cut every possible subset of the set A.

Lemma 7.5. *If H has infinite VC dimension, then H is not PAC-learnable.*

Proof. (Sketch) We will connect to the NFL theorem. The intuition goes as follows. NFL theorem states that you need to observe at least half of the data to learn. Now, VC dimension tells us the size of the dataset we can shatter. If VC dimension is infinite then you need "half of that". So not learnable. □

Lemma 7.6. Sauer's lemma *Let H be a class of binary valued functions, and let $VC_{dim} = d < \infty$ then for every m,*

$$\Pi_H(m) \leq \sum_{i=0}^{d} \binom{m}{i} \leq \left(\frac{em}{d}\right)^d \leq m^d,$$

where,

$$\Pi_H(m) = \max_{|A| \leq m} |\{h \cap A : h \in H\}|,$$

stands for how many subsets can H cut from A. This is the same as

$$= |\{g : A \to \{0,1\} : \exists h \in H, \forall a \in A, g(a) = h(a)\}|.$$

This is important because the number of possible binary colorings of m points is 2^m, and Sauer's lemma tells us that the moment we have a finite VC dimension the colorings that we may have is dramatically reduced to a polynomial number, i.e. $|H| \leq m^{VC_{dim}}$.

7.3 The fundamental theorem of statistical learning

Theorem 7.7. *For every domain X and every class H of functions from X to $\{0,1\}$, the following statements are equivalent:*

1. *H has the uniform convergence property*[1].
2. *ERM is a successful agnostic PAC-learner for H.*
3. *H is agnostic PAC-learnable.*
4. *ERM is a successful PAC-learner for H.*
5. *H is PAC-learnable.*
6. *VC_{dim} is finite ($VC_{dim}(H) < \infty$).*

Theorem 7.8. *There exists constants $C_!$ and C_2 such that for any class H of finite $VC_{dim} = d$, we get the following inequalities*

1. *In the PAC sense (realizable case) $\forall \epsilon, \delta > 0$,*

$$C_1 \frac{d + \ln 1/\delta}{\epsilon} \leq m_H^{PAC}(\epsilon, \delta) \leq C_2 \frac{d \ln 1/\epsilon + \ln 1/\delta}{\epsilon},$$

2. *For Agnostic PAC,*

$$C_1 \frac{d + \ln 1/\delta}{\epsilon^2} < m_H^{AgPAC}(\epsilon, \delta) \leq C_2 \frac{d + \ln 1/\delta}{\epsilon^2},$$

3. *For Uniform convergence (this is what sample size we need to be ϵ-representative)*

$$" < m_H^{UC}(\epsilon, \delta) \leq ",$$

4. *For ERM to be successful,*

$$" < m_H^{ERM}(\epsilon, \delta) \leq ".$$

Observation 5. *Recall the sample complexity of agnostic PAC-learnable in the finite case and compare it to the current bound*

$$m_H^{AgPACfin}(\epsilon, \delta) \leq \frac{2 \log(\frac{2|H|}{\delta})}{\epsilon^2} = 2 \frac{\log(2|\mathcal{H}|) + \log 1/\delta}{\epsilon^2},$$

$$m_H^{AgPAC}(\epsilon, \delta) \leq C_2 \frac{d + \ln 1/\delta}{\epsilon^2}.$$

Observe that d plays the role of the size of the hypothesis class. And recall the bound on d, i.e. $d \leq \log(|H|)$. This means that using d we have a tighter bound and save samples.

[1] For every ϵ, δ we have a sample size that if we take that amount of samples it is guaranteed that it is ϵ-representative